

MAHESH REDDY CHANGAL

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Education

University of Maryland, Baltimore County

Master of Science, Computer Science

August 2023 – May 2025

GPA: 3.7 / 4.0

Experience

Tiger Analytics

Chicago, IL

Software Engineer II – AI/Cloud Solutions

Jan 2026 – Present

- Worked directly with **customer engineering teams** to gather requirements, author **technical design documents**, and jointly develop **cloud-based AI solutions** on **Azure**, translating ambiguous business problems into concrete solution architectures with measurable outcomes
- Architected and co-developed production **agentic AI systems** using LangChain and LangGraph with **Azure AI Search** and **Azure OpenAI**, enabling autonomous multi-step reasoning pipelines that reduced customer manual intervention by 60% across enterprise workflows
- Designed **Retrieval-Augmented Generation (RAG)** pipelines integrating **LLMs** with domain-specific knowledge bases on **Azure Cognitive Services**, improving response accuracy by 40% for business-critical customer applications
- Acted as **Designated Responsible Individual (DRI)** for production AI services, managing on-call rotations to monitor system health, diagnose degradations, and restore service continuity for enterprise customers with SLA requirements
- Broke down complex feature deliverables into sprint-sized work items, provided **effort estimation**, and coordinated delivery across **cross-functional teams** including data science, DevOps, and product – shipping features 30% faster than baseline
- Contributed **open-source** solution patterns and reusable code assets to internal and external repositories, enabling other customer teams to accelerate their own **cloud solution** development on the Microsoft platform

University of Maryland, Baltimore County

Baltimore, MD

Graduate Assistant & Grader

Aug 2024 – May 2025

- Evaluated **Python**, **Java**, and **C++** implementations for 200+ students across Data Science and Network Security courses, conducting structured **code reviews** that reduced recurring errors by 35% and reinforced industry software engineering practices
- Partnered with faculty to **design and document** evaluation rubrics for distributed systems and ML coursework, producing clear technical documentation accessible to both engineering and non-technical audiences
- Mentored students weekly on **data structures**, **algorithms**, and performance optimization, improving average solution efficiency by 25% and building a pipeline of stronger junior engineers

Tiger Analytics

Hyderabad, India

Senior Software Engineer (promoted from Data Engineer)

Nov 2021 – July 2023

- Partnered with **customer stakeholders** to define user requirements and architect a no-code **Terraform** automation platform using **React** (TypeScript) and Flask/MongoDB, empowering 500+ users to provision **cloud infrastructure** visually and cutting deployment time by 40%
- Wrote and maintained **design documents** for a scalable web scraping and metadata pipeline (Selenium, BeautifulSoup) that exposed normalized **REST APIs** across **AWS**, **Azure**, and **GCP** service catalogs – reducing platform maintenance overhead by 50%
- Led end-to-end migration of critical ETL pipelines from **Azure** to **GCP**, rebuilding data workflows in Databricks with Spark – achieving 25% cost reduction, 99.9% uptime, and zero data loss
- Developed **large-scale data processing** pipelines using Apache Airflow to orchestrate PySpark jobs on **AWS EMR**, processing 5TB+ of daily data with automated scheduling, cutting processing time by 20%
- Implemented **CI/CD pipelines** and led **agile** sprint planning, decomposing roadmap initiatives into sized work items and collaborating with cross-functional teams (PM, DevOps, UX) – accelerating release cycles by 30% and driving 25% increase in platform adoption

ICRISAT

Hyderabad, India

Research Intern

June 2019 – July 2019

- Developed a **machine learning model** using TensorFlow, Pandas, and NumPy to predict optimal fertilizer rates, improving agricultural efficiency by 15% and delivering data-driven insights to research stakeholders
- Built and deployed a customer-facing internal platform using Django and **React** on **AWS**, providing real-time updates to 200+ field researchers and demonstrating end-to-end **cloud solution** delivery from requirements to launch

Projects

PaperMind – RAG Document Intelligence Platform | *Python, FastAPI, React, Azure Cognitive Search, LangChain, ChromaDB*

- Designed and built a full-stack **RAG** application end-to-end – from requirements and **design documents** through deployment – featuring PDF ingestion, LangChain chunking, **vector storage**, and contextual Q&A powered by open-source LLMs; serves as a reusable **open-source** solution pattern for document intelligence
- Architected a summary-first retrieval layer with embedding pipelines (sentence-transformers), improving answer relevance by 35% and demonstrating the same **hybrid search** mechanics used in production **Azure AI Search** deployments

Distributed File System (GFS-inspired) | *Python, gRPC, Docker, Distributed Systems*

- Architected a fault-tolerant distributed file system using Python and **gRPC**, implementing leader election and replication protocols that achieved 99.9% availability and zero data loss during node failures – directly applicable to **cloud solution** reliability patterns
- Containerized all services with **Docker**, implemented heartbeat monitoring and automated failover, and optimized inter-service communication to reduce latency by 50% – illustrating production-grade **observability and reliability** practices

Conversational AI Agent Platform | *TypeScript, React, Node.js, LangChain, OpenAI API, WebSockets*

- Built a production-grade conversational agent with tool-use capabilities and multi-step reasoning, integrating OpenAI via LangChain with streaming responses and conversation memory – contributes to **open-source** patterns for agentic **cloud solutions**
- Implemented real-time WebSocket architecture supporting concurrent sessions, prompt engineering pipelines with token management, and end-to-end **model evaluation** to ensure consistent quality across diverse conversational scenarios

Technical Skills

Programming Languages: Python, Java, TypeScript, JavaScript, C#, Go, SQL

Cloud Platforms: Azure (AI Search, OpenAI, Cognitive Services, Functions, Blob, Synapse), AWS (EC2, S3, Lambda, SQS, EMR), GCP, Terraform, Kubernetes, Docker, CI/CD Pipelines

AI/ML & Generative AI: LangChain, LangGraph, RAG Pipelines, Agentic Systems, LLMs, Azure OpenAI, Embeddings, Semantic Search, Vector Databases (Pinecone, Weaviate, ChromaDB, FAISS), NLP, TensorFlow, PyTorch

Frontend & Backend: React, Next.js, Node.js, Spring Boot, Django, Flask, FastAPI, REST APIs, GraphQL, Microservices, WebSockets

Data Engineering: Apache Spark, PySpark, Apache Airflow, ETL Pipelines, Databricks, Delta Lakes, Azure Synapse

Databases: PostgreSQL, MongoDB, MySQL, Redis, Elasticsearch

Engineering Practices: Agile, Design Reviews, Code Reviews, Technical Documentation, System Design, On-Call/DRI, Cross-Functional Collaboration, Open Source, TDD, Git